To Whom It May Concern,

I am pleased to write this letter of recommendation for Farhad Shadmand, who worked with me as a research intern at Adobe Research in the summer of 2025. During his three-month internship, Farhad led a challenging project on robust print watermarking, culminating in a CVPR submission.

The goal of the project was ambitious: design a watermarking system that survives real-world print-capture pipelines while preserving visual fidelity. To that end, we decided to attack the problem from two fronts simultaneously:
(1) improving the network architecture for watermark encoding and decoding, and
(2) building a realistic print noise simulation pipeline that could model color and geometric distortions introduced by printers and scanners.

This is a lot to expect from a three-month internship, yet Farhad exceeded these expectations. Technically, Farhad is extremely strong. He quickly familiarized himself with internal training platform and then proceeded to design, implement, and evaluate several variants of the print watermarking model, including the rotation invariant encoder–decoder architecture and the new noise simulation network that are central contributions of the paper.

He independently explored diffusion models and GAN-based approaches for print simulation and integrated the best-performing variant into the watermark training loop as a drop-in augmentation module. His experiments were always well-designed, carefully logged, and easy for the rest of the team to reproduce.

One moment that stands out to me illustrates both his persistence and maturity as a researcher. Midway through the internship, we realized that directly using a diffusion model inside the training loop was too slow and unstable to be practical. Instead of getting discouraged, Farhad proposed a two-stage solution: first using the diffusion model offline to generate a large synthetic print–scan dataset and then training lightweight GAN models as our Noise Simulation Network. Within a week he had a working prototype, and within a few more iterations we saw clear quantitative and qualitative gains over traditional noise augmentations. That pivot ended up being a key part of our final system.

Beyond his technical contributions, Farhad was a model intern. He was disciplined and reliable, always prepared for our regular meetings with fresh results, plots, and concrete next steps. He presented his work several times to a larger cross-site team at Adobe, handling detailed technical questions with clarity and humility. He also drove the drafting of our internal invention disclosure for patent filing and contributed substantially to the writing of the paper - both the methodological sections and the experimental analysis.

In a single summer, Farhad managed to:

- contribute to a state-of-the-art print watermarking architecture,
- design and deploy a novel print noise simulation pipeline,
- first-author a CVPR submission, and

- help prepare an internal patent disclosure.

This output would be impressive even for a full-time researcher; for a three-month intern, it speaks to his work ethic, creativity, and ability to deliver under time constraints.

In summary, I recommend Farhad **strongly and without reservation** for any research-oriented position. He combines strong technical skills with genuine curiosity, persistence in the face of setbacks, and the communication skills needed to thrive in collaborative research environments. I would be very happy to work with him again.

Please feel free to contact me if you require any additional information.

Sincerely,

Shruti Agarwal
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